UNIVERSITY OF CAMBRIDGE INTERNATIONAL EXAMINATIONS

GCE O Level

MARK SCHEME for the May/June 2006 question paper

5054 PHYSICS

5054/04 **Paper 4** **maximum raw mark 30**

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1 **(a)** **(i)** length clearly marked, from top or bottom of rings

(ii) vertical ruler drawn within ½ cm of load

(iii) eye/observer positioned to avoid parallax **[3]**

(b) values in table in ascending/descending order **[1]**

(c) axes: – correct way round, labelled quantity and unit
scales: – more than ½ grid, sensible
points: plotted accurately (within ½ square) and neat
line: straight line best fit drawn with ruler, neat **[4]**

(d) **(i)** spring has length with no load

(ii) no line does not go through origin **[2]**

(e) **(i)** increase in length

(ii) 20.5 $\pm$ 0.5 cm

(iii) straight line through origin **[4]**

[Total: 14]

2 **(a)** iron/soft iron/mumetal **[1]**

(b) any suitable method which will give a comparison
e.g. how many/mass paper clips/pins/nails/tacks holds,
distance from paper clip to make paper clip move/jump
distance from compass to make it move **[1]**

(c) A: does not change circuit/current/resistance **[1]**

[Total: 3]

3 **(a)** connects battery, bulb, component in series **[1]**

(b) reverses connections in box/battery
checks brightness **[2]**

(c) no light
bright both ways
bright one way, off when connections reversed
dim both ways **[4]**

[Total: 7]

4 **(a)** 0.8 $\pm$ 0.1 cm 5.0 $\pm$ 0.1 cm **[1]**
draw parallel tangents/measure more than one in different places **[1]**

(b) 7.85 cm^3 ecf (one, two or 3sf) **[1]**

(c) thickest part in the centre/at distance from ruler/
parallax error explained **[1]**

(d) **(i)** displacement of water described,
volume displaced equals volume of lens

(ii) volume of water displaced small, needs large displacement can/
measuring cylinder/large scale on measuring cylinder **[2]**

[Total: 6]

[Paper Total: 30]

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